Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 11

Deadline: next Friday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week.

Please submit written solutions to **TWO** of the following problems.

Problem 1. Give an example of cyclic groups G and H such that $G \times H$ is not cyclic.

Problem 2. Use the classification of groups with six elements to show that A_4 has no subgroup with 6 elements.

Problem 3. Consider the binary code

$$\mathcal{C} = \{\texttt{00000}, \texttt{11100}, \texttt{00111}, \texttt{11011}\} \subseteq \mathbb{B}^5.$$

Recall that the distance between two words is the number of positions where they differ.

(a) Compute all pairwise distances between distinct codewords in $\mathcal{C}$.

(b) Find the minimum distance $d_{\min}$ of this code.

(c) Using the error-detection theorem, how many errors is this code guaranteed to detect?

(d) Give an example of a one-error received word that would be detected if `00111` were sent.

(e) Give an example of a three-error change that sends one codeword in $\mathcal{C}$ to another codeword in $\mathcal{C}$. Explain why this kind of error would not be detected.

Problem 4. For the parity check code $f : \mathbb{B}^4 \to \mathbb{B}^5$, the last bit is chosen so that the codeword has an even number of 1s.

(a) Compute $f(\texttt{1011})$ and $f(\texttt{0100})$.

(b) Explain why this parity check code detects every odd number of bit flips.

(c) Give an example of a two-error change that is not detected by the parity check code.